MAXWELL HIGDON

Senior Hardware Engineer

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PROFESSIONAL OVERVIEW

Mechanical Engineer with expertise in hardware design, system integration, and global field support. Currently a Senior Engineer at Ricoh USA, leading cross-functional teams on deployment and enhancement of Continuous Form Inkjet systems. Skilled in system-level redesigns, troubleshooting, and technical advisement. Brings hands-on experience in CAD, thermal analysis, and manufacturing, with a project background in assistive technologies and cooling energy systems.

EXPERIENCE

Senior Engineer, Hardware

Ricoh USA

April 2025 – Ongoing Boulder, CO

- Led cross-functional engineering efforts to implement site-specific modifications for the Ricoh Pro VC80000, improving system reliability and deployment efficiency across multiple installations
- Designed and implemented upgraded cooling fluid transport systems, optimizing flow distribution and thermal performance.
- Diagnosed and resolved complex electromechanical system failures in live production environments, minimizing downtime.

Associate Engineer, Hardware

Ricoh USA

July 2023 – April 2025 Boulder, CO

- Delivered technical support for global installations and maintenance of continuous form inkjet platforms across customer sites.
- Updated and revised engineering drawings to support custom component fabrication and field-driven design changes.
- Collaborated with senior engineers to troubleshoot system-level issues spanning mechanical, fluid, and control systems.

Student Intern II, MEP

Olsson

January 2023 – May 2023 Denver, CO

- Modeled HVAC and plumbing systems in Revit and AutoCAD for hospitals and commercial buildings, ensuring compliance with project and code requirements.
- Developed internal reference standards to improve consistency and efficiency across Trace3D, Revit, and AutoCAD workflows.
- Assisted in system layout design and coordination with multidisciplinary teams to support construction documentation.

Engineering Intern

Cole Garner Geotechnical

May 2021 – August 2021 Westminster, CO

- Researched geothermal pile heat exchanger systems, analyzing ground temperature interactions to enable passive cooling systems.
- Developed a predictive modeling tool to estimate temperature change and heat exchange performance across various environmental conditions.

EDUCATION

B.S. Mechanical Engineering

University of Colorado, Boulder

August 2019 – May 2023

SKILLS

- **CAD:** SolidWorks, AutoCAD, Revit.
- **Hardware:** Electromechanical systems, fluid transport, HVAC/plumbing integration.
- **Analysis:** Thermal systems, fluid systems, system-level troubleshooting.
- **Engineering:** GD&T, system integration, root cause analysis.

ACHIEVMENTS

- **Certified SolidWorks Associate**
  Dassault Systèmes - 2020
- **Recognition Award for Onsite Support Excellence**
  Ricoh USA - 2025
- **Recognition Award for Business Development**
  Ricoh USA - 2025

PROJECTS

- **Wheelchair-Compatible Motorized Disc Launcher**
  QL Plus
  2022-2023
- **Assistive Audio Regulation Tool**
  University of Colorado, Boulder
  2019-2020